

Z E N I S T H A I

Zenisth Voice

Product Requirements Document

AI Voice Automation Platform

Version 1.0.0	Status Active Development
Company Zenisth AI	Date February 2026

1. Executive Summary

Zenisth Voice is an AI-powered voice automation platform built for modern sales and support teams. It enables intelligent outbound calling, inbound support handling, behavioral scheduling, human handoff, and full conversation compliance — all from a single unified dashboard.

Unlike generic calling bots, Zenisth Voice is designed around human psychology and enterprise trust. Every call is time-aware, behavior-responsive, evidence-backed, and workflow-driven.

Core Value Proposition

Pillar	Description
Voice Automation	AI agents handle outbound & inbound calls end-to-end
Behavioral Intelligence	System respects timing, intent, and human preferences
Workflow Execution	Calls trigger CRM updates, emails, bookings automatically
Monitoring & Proof	Full audio, transcript, and source-linked evidence per call
Enterprise Governance	Compliance, role access, escalation, and audit layers

Product Goal

Enable any business to scale voice-based outreach and support responsibly — without sacrificing trust, accuracy, or human oversight.

2. Product Scope & Boundaries

2.1 In Scope (MVP)

- AI Outbound Calling Agent with campaign management
- AI Inbound Support Agent with SOP-based answering
- Behavioral Scheduling Engine (callback, time-zone, no-call rules)
- Human Handoff Console with context summary
- Knowledge Center (upload SOP, docs, FAQs)
- Conversation Timeline (audio + transcript + source)
- Compliance & Dispute Layer
- Agent Builder (no-code voice agent configurator)
- Basic CRM webhook integration (HubSpot, Salesforce)
- Dashboard with real-time monitoring

2.2 Out of Scope (V2+)

- Multi-language AI voice (V2)
- Video call support (V3)
- Native mobile app (V2)
- Custom LLM fine-tuning per account (V2)
- Zenisth Transify integration (V2)
- FlowraAI deep workflow orchestration (V2)

i MVP focus is voice automation + proof + human oversight. All advanced integrations are planned for V2.

3. User Personas

Persona	Role	Primary Goal	Pain Point
Alex	Sales Manager	Scale outbound outreach without hiring more reps	Reps waste time on no-answer calls & manual follow-ups
Priya	Support Lead	Automate tier-1 support queries 24/7	Agents burn out on repetitive FAQ calls
Rohan	Founder / Operator	Get full control & proof of every AI conversation	Fear of AI saying wrong things to customers
Neha	Compliance Officer	Ensure every call is logged, auditable, and safe	No audit trail for AI-generated calls
Sam	SDR / Sales Rep	Get notified for warm leads & callbacks	Missing follow-up windows on interested prospects

4. Information Architecture & Site Map

4.1 Navigation Structure

Zenisth Voice is structured as a single-product SaaS platform with a left sidebar navigation. The primary nav is organized into 6 zones:

Zone	Module	Access Level
Home	Dashboard Overview	All Roles
Knowledge	Knowledge Center	Admin, Manager
Agents	Agent Builder	Admin, Manager
Campaigns	Campaign Manager	Admin, Manager, Rep
Conversations	Conversation Timeline	All Roles
Monitor	Live Monitoring Console	Admin, Manager
Humans	Human Handoff Console	Rep, Manager
Compliance	Audit & Compliance Layer	Admin, Compliance Officer
Settings	Account, Integrations, Billing	Admin Only

4.2 Screen Flow Map

Outbound Campaign Flow

1. Login → Dashboard
2. Dashboard → Campaigns → Create Campaign
3. Create Campaign → Upload Leads → Assign Agent → Set Schedule → Launch
4. Campaign Running → Monitor View (Live)
5. Call Ends → Conversation Timeline → View Transcript + Audio
6. Lead Interested → Human Handoff Console (auto-alert to rep)
7. Callback Requested → Scheduler creates callback entry → Rep notified or AI auto-calls

Inbound Support Flow

8. Inbound Call Received → AI Agent picks up
9. AI checks Knowledge Center → Answers from SOP

10. If confidence < threshold → Escalation triggered
11. Human Console shows live call with context
12. Rep takes over → Call continues
13. Call ends → Full log saved to Conversation Timeline

5. Feature Modules — Detailed Specifications

5.1 Dashboard Overview

Purpose

Command center for all voice activity. First screen after login. Shows real-time metrics, active campaigns, live calls, and pending actions.

Key Widgets

Widget	Data Shown	Update Frequency
Active Calls	Count of live AI calls in progress	Real-time (WebSocket)
Calls Today	Total calls made today	Every 30 seconds
Answer Rate	Calls answered / calls attempted (%)	Every 30 seconds
Conversion Rate	Leads converted / total calls (%)	Every 30 seconds
Pending Callbacks	Count of scheduled callbacks due today	Real-time
Handoff Queue	Calls waiting for human takeover	Real-time
Campaign Status	Running / Paused / Completed badges	On event
Recent Activity Feed	Last 10 call events with status	Every 60 seconds

Design Rules

- 4px grid system throughout. All spacing in multiples of 4px.
- Primary brand color #5A46C7 for active states and key metrics.
- Red badges only for critical alerts (handoff queue, failed calls).
- Cards use 8px border-radius, 1px border in #C8C8D8.
- Nunito font: 14px body, 16px labels, 20px metric values, 24px widget titles.

5.2 Knowledge Center

Purpose

The single source of truth for the AI agent. All answers the AI gives during a call must originate from documents uploaded here. No hallucination is permitted — if the AI cannot find the answer in the knowledge base, it must escalate.

Document Types Supported

- PDF (SOPs, brochures, product manuals)
- DOCX (internal documentation)
- TXT (FAQs, scripts)
- CSV (product catalog, pricing tables)
- URL import (website pages parsed and indexed)

Core Features

Feature	Description
Document Upload	Drag-and-drop or file picker. Max 50MB per file. Multiple files supported.
Auto-Chunking	System splits docs into semantic chunks for vector search.
Version Control	Each upload creates a versioned snapshot. Agents always use latest unless pinned.
Doc Status	Pending Index → Processing → Active → Archived
Test Query	Type a question and see what chunk the AI would retrieve — before going live.
Source Tagging	Each doc is tagged (SOP, Pricing, FAQ, Legal) for filtered retrieval.
Access Control	Admin can restrict which agents access which documents.

Edge Cases — Knowledge Center

⚠ Edge Case 1: Document upload fails due to file size or unsupported format. System shows specific error message: "File exceeds 50MB limit" or "Format not supported." Retry option provided. No silent failures.

⚠ Edge Case 2: Document processing takes longer than 2 minutes (large PDF). UI shows "Processing..." state with live progress. Agent cannot be activated until all knowledge docs are Active.

✗ Edge Case 3: Conflicting information across two documents. System surfaces conflict warning on test query. Admin must resolve by archiving outdated document or marking a priority source.

✗ Edge Case 4: Document deleted while agent is live. System freezes knowledge base for active sessions. Change applied only to new sessions. Admin is notified of the dependency.

5.3 Agent Builder

Purpose

No-code interface to design and configure an AI voice agent. Defines the agent's persona, tone, call flow, escalation rules, and knowledge source.

Agent Configuration Fields

Field	Type	Description
Agent Name	Text Input	Internal label. Shown in Monitoring view.
Voice Persona	Dropdown	Select from prebuilt voice profiles (Friendly, Professional, Assertive)
Language	Dropdown	English (MVP). Multi-language in V2.
Opening Script	Rich Text	Exact intro the AI says on call connect. Variables: {contact_name}, {company_name}
Objective	Dropdown	Outbound Sales / Inbound Support / Appointment Setting / Survey
Knowledge Source	Multi-select	Attach documents from Knowledge Center
Confidence Threshold	Slider (0–100%)	Below this % → escalate to human. Default: 75%.
Escalation Rule	Rule Builder	Define when to hand off: keyword match, confidence, repeat question, user request
Max Call Duration	Number Input	Hard limit in minutes. AI ends call gracefully if exceeded.
Callback Capture	Toggle	Enables AI to detect and log callback requests from users.
Ending Script	Rich Text	Closing statement. Always plays before call disconnect.

Call Flow Designer (Visual)

A node-based flow canvas lets admins map the call conversation tree. Nodes include:

- START — Opening statement
- LISTEN — AI listens for user intent
- RETRIEVE — Knowledge lookup from uploaded docs
- RESPOND — AI speaks the answer
- DECISION — Branch based on user response (Yes/No/Other)
- SCHEDULE — Capture appointment or callback time
- HANDOFF — Transfer call to human with summary
- END — Closing script and call disconnect

Edge Cases — Agent Builder

X Edge Case 1: Admin tries to activate an agent with no knowledge source attached. System blocks activation. Error: "Agent requires at least one active knowledge document."

⚠ Edge Case 2: Admin sets confidence threshold to 0%. System warns: "Setting threshold to 0% disables escalation. All uncertain answers will be delivered. Are you sure?" Requires explicit confirmation.

X Edge Case 3: Opening script contains an undefined variable like {undefined_field}. System highlights the broken variable in red and blocks save until resolved.

i Edge Case 4: Agent is edited while a live campaign is running. Changes are queued and applied only after current campaign session ends. Banner shown: "Changes will apply after active campaign completes."

5.4 Campaign Manager

Purpose

Create, launch, and manage outbound calling campaigns. Upload lead lists, assign agents, configure timing rules, and track real-time progress.

Campaign Creation Steps

- 14. Name the Campaign — Internal label + optional description.
- 15. Upload Lead List — CSV with required fields: name, phone, company (optional), timezone.
- 16. Assign Agent — Select a configured AI agent from Agent Builder.
- 17. Set Calling Schedule — Define daily windows (e.g., 9 AM–6 PM local time). Select active days.
- 18. Configure Retry Rules — Max attempts per lead, gap between retries (min 4 hours recommended).
- 19. Set Campaign Goal — Demo booking / Information delivery / Survey / Lead qualification.
- 20. Review & Launch — Summary screen before going live.

Campaign Status States

Status	Description	Allowed Actions
Draft	Campaign created but not launched	Edit, Launch, Delete
Scheduled	Launch time is in future	Edit schedule, Cancel
Running	Calls actively being made	Pause, Monitor, View
Paused	Temporarily stopped by admin	Resume, Edit, Archive
Completed	All leads contacted or daily limit reached	Clone, View Report
Archived	Manually archived by admin	Restore, View
Failed	Launch error (e.g., no credits, invalid leads)	View Error, Retry

Behavioral Scheduling Engine

The Behavioral Scheduling Engine is a core differentiator. It governs when and how the AI calls leads:

- Time-Zone Awareness: Reads lead timezone from CSV. Never calls outside the configured safe window in the lead's local time.

- Callback Capture: If a lead says "Call me at 6 PM" or similar phrase, the system captures the time, creates a callback entry, and either alerts the assigned rep or schedules an auto-call.
- Retry Throttle: Minimum 4-hour gap between retry attempts on the same lead. Max 3 attempts default (configurable).
- DNC Compliance: Leads marked as "Do Not Call" are permanently suppressed from all future campaigns.
- Engagement-Based Stop: If a lead has not answered after X attempts (admin-defined), the system auto-pauses further outreach to that lead.

Edge Cases — Campaign Manager

⚠ Edge Case 1: CSV upload has duplicate phone numbers. System deduplicates automatically and shows count: "3 duplicate entries removed." Admin can download the duplicate report.

✗ Edge Case 2: Campaign launched but account has insufficient call credits. System blocks launch. Error: "Insufficient credits for this campaign size. Estimated cost: X credits. Available: Y credits." Direct link to billing.

i Edge Case 3: All leads in a campaign are in a timezone where it is currently outside calling hours. Campaign enters "Waiting for Local Window" state — not failed, not paused. Resumes automatically.

i Edge Case 4: Lead answers call but immediately hangs up. System logs as "Rejected" not "Answered." Retry rules still apply. Rejection is factored into engagement scoring.

✗ Edge Case 5: A lead asks the AI to stop calling. AI responds: "Absolutely, I'll make sure you're removed from our list." System auto-adds the lead to DNC list immediately. No retry ever.

5.5 Conversation Timeline

Purpose

Every call is permanently recorded, transcribed, and linked to the source document the AI used. This is the proof and compliance layer of the platform.

Call Record Components

Component	Description
Call Metadata	Date, time, duration, agent used, campaign name, lead info
Audio Playback	Full call recording with waveform visualization and speed control
Transcript	Verbatim text with speaker labels (AI / Human) and timestamps per line
Source Citations	Each AI response is linked to the exact document chunk it referenced
Outcome Tag	Meeting Booked / Callback Requested / Not Interested / DNC / Escalated / Error
Confidence Scores	Per-response confidence shown as color-coded bars (Green: high, Orange: mid, Red: low)
Escalation Log	If handoff happened: timestamp, reason, rep who took over
Notes	Rep or manager can add manual annotations to any call record

Filters & Search

- Filter by: Campaign, Agent, Date Range, Outcome, Escalated, Duration
- Full-text search across all transcripts
- Sort by: Date, Duration, Confidence Score, Outcome

Edge Cases — Conversation Timeline

X Edge Case 1: Audio recording fails due to connectivity issue mid-call. Transcript is still saved. Record is flagged: "Audio unavailable — transcript only." Compliance officer is notified.

⚠ Edge Case 2: Transcript generation fails (e.g., heavy accent or background noise). Record is flagged: "Transcript pending review." Audio is always available as fallback.

✗ Edge Case 3: Admin attempts to delete a call record. System blocks deletion if record is less than 90 days old (configurable retention policy). Only Compliance Officer role can override with reason logged.

5.6 Live Monitoring Console

Purpose

Real-time view of all active calls in progress. Managers can listen silently, see live transcript, check AI confidence in real time, and intervene if needed.

Features

Feature	Description
Active Call Grid	Card-based view of all live calls. Cards show: lead name, agent name, call duration, current stage.
Live Transcript Feed	Click any call to expand and read the transcript as it's being generated.
Confidence Indicator	Color ring on each card: Green (confident), Orange (borderline), Red (about to escalate).
Silent Listen Mode	Manager can listen to live audio without being heard on the call.
Force Escalate Button	Manager can instantly push a call to human handoff from this view.
Campaign Filter	View only calls from a specific campaign.
Alert Panel	Right panel shows: calls nearing max duration, confidence drops, escalation events.

Edge Cases — Monitoring Console

- ⚠ Edge Case 1: Manager clicks "Force Escalate" on a call that was already escalated by the AI simultaneously. System prevents double-escalation. One handoff is created. No conflict in Human Console.
- ℹ Edge Case 2: More than 50 simultaneous active calls. UI switches to compact list view automatically. Individual card expansion still works. Performance optimized via virtualized rendering.

5.7 Human Handoff Console

Purpose

When the AI cannot handle a call — due to low confidence, user request, keyword trigger, or manager override — the call is transferred to a human rep. This console manages that transfer with full context.

Handoff Flow

- 21. Escalation triggered (by AI, by user request, or by manager).
- 22. AI says to caller: "Let me connect you with a specialist who can help you better."
- 23. AI does NOT drop the call — it stays on hold music for up to 2 minutes.
- 24. Human Console shows: alert notification, call context card with full summary.
- 25. Context Card includes: lead name, company, conversation summary, AI's last answer, reason for escalation.
- 26. Rep clicks "Accept" → Call bridges to rep's phone or web dial.
- 27. Rep completes the call → Marks outcome → Call logged with handoff record.

Context Card Contents

Section	Content
Lead Info	Name, phone, company, timezone, campaign name
Conversation Summary	AI-generated 3-5 sentence summary of what was discussed
Last AI Response	The exact last thing the AI said before escalation
Escalation Reason	Confidence too low / User said "speak to human" / Keyword match / Manager override
Recommended Next Step	AI-suggested action: "Lead asked about enterprise pricing. Offer a custom quote."

Edge Cases — Human Handoff Console

X

Edge Case 1: No rep is available to accept the handoff. AI holds for 2 minutes then says: "Our team is busy right now. May I schedule a callback for you?" If yes — callback created. If no — call ends gracefully and lead is flagged for priority follow-up.

⚠ Edge Case 2: Rep accepts the handoff but has a poor connection. System detects audio quality drop and prompts rep: "Poor connection detected. Switch to browser dial?" Does not drop caller.

i Edge Case 3: Caller hangs up during handoff hold music. System marks call as "Dropped During Transfer." Lead is automatically scheduled for a priority callback. No permanent DNC applied.

5.8 Compliance & Audit Layer

Purpose

Enterprise-grade evidence system. Every AI action is logged, sourced, and attributable. Designed for regulated industries, dispute resolution, and internal audits.

Compliance Features

Feature	Description
Immutable Call Logs	All call records are write-once. Cannot be modified after save.
Source Attribution	Every AI statement links to the exact knowledge document chunk used.
Audit Trail	Every config change (agent edits, knowledge updates, DNC changes) is logged with user, timestamp, and before/after state.
DNC Registry	Centralized do-not-call list. Syncs in real time across all campaigns.
Call Consent Verification	System logs that calling complied with configured timezone windows and consent rules.
Export Reports	Full audit reports exportable as PDF or CSV for any date range.
Role-Based Access	Compliance Officer has read-only access to everything. Cannot modify but can flag.
Data Retention Policy	Admin configures retention window (30/60/90/180 days). Auto-archive after retention.

Edge Cases — Compliance

X Edge Case 1: Admin attempts to edit an audit log entry. System blocks all edit attempts. Error: "Audit logs are immutable and cannot be modified." Only deletion by Super Admin after retention period is allowed.

i Edge Case 2: A lead disputes that the AI gave incorrect information. Compliance officer can pull: the exact call audio, verbatim transcript, and the source document chunk the AI cited. Full proof chain available in one screen.

6. End-to-End Workflows

6.1 Workflow A: Outbound Cold Call Campaign

Actor: Sales Manager, AI Agent, Lead

Step	Who	Action	System Response
1	Manager	Creates campaign, uploads 500 leads CSV	System validates CSV, deduplicates, geocodes timezones
2	Manager	Assigns AI Agent, sets calling window 9AM-6PM	Campaign enters Draft state
3	Manager	Clicks Launch	Campaign moves to Running. Scheduler starts queuing calls.
4	AI Agent	Calls Lead #1	Lead does not answer. Logged as "No Answer." Retry after 4 hours.
5	AI Agent	Calls Lead #2. Lead answers.	AI delivers opening script with lead name variable.
6	Lead	"Tell me more about your pricing"	AI retrieves from pricing doc. Responds with sourced answer.
7	Lead	"Can you call me back at 6 PM?"	AI confirms. System creates callback. Rep is notified.
8	AI Agent	Closing script played. Call ends.	Full record saved: audio + transcript + source + outcome = Callback Scheduled
9	System	At 5:55 PM — reminder to rep	Rep gets notification: "Callback due at 6 PM for Lead #2"
10	AI / Rep	Callback at 6 PM	Rep or AI calls. If rep: context card shown. If AI: previous convo context loaded.

6.2 Workflow B: Inbound Support Call

Step	Who	Action	System Response
1	Customer	Calls support number	AI Agent answers within 2 rings. Greeting plays.
2	Customer	"My order hasn't arrived after 10 days"	AI retrieves shipping SOP. Asks for order number.
3	Customer	Provides order ID	AI checks integrated order lookup (if configured) or provides SOP-based guidance.
4	Customer	"I want a refund now"	Keyword "refund" triggers escalation rule. Confidence check run.
5	AI Agent	Confidence 42% — below threshold	Escalation triggered. AI: "Let me connect you with a specialist."
6	System	Handoff initiated	Human Console shows alert. Context card populated with summary.
7	Rep	Accepts handoff within 90 seconds	Call bridges. Rep sees full context. Resolves refund.
8	System	Call ends	Record saved: audio + transcript + escalation log + rep outcome tag

6.3 Workflow C: Callback Scheduling

Step	Who	Action	System Response
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1	AI Agent	On a call. Lead says "Call me at 6 PM tomorrow"	NLP detects callback intent. Extracts: time=6PM, day=tomorrow.
2	AI Agent	Confirms with lead: "I'll schedule a callback for tomorrow at 6 PM your time. Is that right?"	Waits for confirmation.
3	Lead	"Yes, that works."	Callback entry created in system. Time converted to UTC.
4	System	Routes callback	If rep assigned: notification sent. If AI callback: queued in scheduler.
5	System	15 minutes before callback	Rep reminder notification. Pre-call context card loaded.
6	Rep / AI	Calls lead at exact time	System confirms timezone compliance before dialing.
7	Lead	Answers or does not answer	Answered: conversation continues. No answer: one retry after 30 mins. Then flagged.

7. UX & Design System

7.1 Design Principles

- Clarity over Cleverness: Every screen should communicate one primary action.
- Trust-First Interfaces: Show evidence, sources, and status at all times.
- Progressive Disclosure: Advanced settings are collapsed by default. Power users can expand.
- Error Prevention Over Error Correction: Guide users before mistakes happen.
- Consistent Feedback: Every action (save, launch, delete) has a visible success or error state.

7.2 Brand & Design Tokens

Color Palette

Token	Hex Value	Usage
Primary	#5A46C7	Buttons, links, active states, metric highlights
Primary Light	#7B6CD4	Hover states, secondary accents
Primary Dark	#3D2FA8	Pressed states, sidebar active
Accent BG	#F4F2FF	Card backgrounds, section fills
Accent Deep	#E8E4FF	Hover fills, tag backgrounds
Neutral 900	#1A1A2E	Primary text, headings
Neutral 700	#2D2D3A	Body text
Neutral 500	#6B6B80	Secondary text, captions
Neutral 200	#C8C8D8	Borders, dividers
Neutral 50	#F5F5FA	Page background, table alternates
Success	#22C55E	Positive outcomes, high confidence
Warning	#F97316	Mid confidence, attention needed
Error	#EF4444	Failed states, low confidence, critical alerts

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Info Teal	#14B8A6	Informational badges, neutral actions	

Grid System

Base unit: 4px. All spacing, padding, margin, and component sizes are multiples of 4px.

Spacing Token	Value	Usage
xs	4px	Icon gaps, fine-tuning
sm	8px	Inline element gaps, tight padding
md	16px	Card padding, field spacing
lg	24px	Section spacing, component gaps
xl	32px	Major section gaps
2xl	48px	Page-level spacing
3xl	64px	Hero sections, cover spacing

Typography — Nunito Font

Role	Size	Weight	Usage
Display	32px	800 (ExtraBold)	Page titles, hero headlines
Heading 1	24px	700 (Bold)	Section headings
Heading 2	20px	700 (Bold)	Sub-section headings
Heading 3	16px	600 (SemiBold)	Card titles, widget headers
Body Large	16px	400 (Regular)	Primary body text
Body	14px	400 (Regular)	General body text
Caption	12px	400 (Regular)	Labels, help text, timestamps
Micro	10px	400 (Regular)	Legal text, footnotes

Component Specs

Component	Spec
Button (Primary)	Height: 40px, Padding: 0 16px, Border-radius: 8px, BG: #5A46C7, Font: 14px/600/Nunito
Button (Secondary)	Height: 40px, Border: 1px #5A46C7, BG: white, Text: #5A46C7
Input Field	Height: 40px, Border-radius: 8px, Border: 1px #C8C8D8, Focus: 2px ring #5A46C7
Card	Border-radius: 12px, Border: 1px #C8C8D8, BG: white, Padding: 24px
Badge/Tag	Height: 24px, Border-radius: 100px, Padding: 0 10px, Font: 12px/600
Sidebar Width	240px collapsed, 64px icon-only mode
Top Nav Height	56px
Table Row Height	48px minimum
Modal Width	480px (small), 720px (medium), 960px (large)

8. Technical Architecture Overview

8.1 System Components

Layer	Technology / Service	Purpose
Frontend	React + TypeScript + TailwindCSS	Web dashboard UI
Voice Infra	Twilio / Vonage / Telnyx	Call routing, SIP, PSTN bridging
Speech-to-Text	AudixAI / Deepgram / Whisper	Real-time transcription
Text-to-Speech	ElevenLabs / Azure TTS	AI voice generation
LLM Layer	GPT-4o / Claude / Llama (configurable)	Intent understanding, response generation
Vector DB	Pinecone / Weaviate	Knowledge center semantic search
Scheduling Engine	Custom Python service + Redis queue	Callback and campaign timing
Storage	S3 / GCS	Call recordings, documents
Database	PostgreSQL	Structured data: leads, campaigns, logs
Realtime	WebSockets (Socket.io)	Live monitoring, handoff alerts
Auth	Auth0 / Supabase Auth	SSO, role-based access
Webhooks	Custom webhook engine	CRM triggers, Zapier, n8n

8.2 Data Flow — Outbound Call

28. Campaign scheduler triggers call at correct local time.
29. Twilio initiates PSTN call to lead.
30. Lead answers → Audio stream opened to server.
31. AudixAI transcribes speech in real-time.
32. Transcript chunk sent to LLM with context: [system prompt + knowledge retrieved from Pinecone + conversation history].
33. LLM generates response text.
34. TTS converts response to audio → streamed back to call.
35. All events logged to PostgreSQL. Audio saved to S3.

36. Call ends → Post-processing: full transcript compiled, source citations mapped, outcome tagged.

9. Integrations

9.1 MVP Integrations

Integration	Type	Direction	Use Case
HubSpot	CRM	Bi-directional	Sync leads, push call outcomes, book meetings
Salesforce	CRM	Bi-directional	Enterprise lead and contact sync
Google Calendar	Calendar	Write	Book meetings from AI conversations
Calendly	Scheduling	Write	AI books slots in rep's Calendly
Zapier	Automation	Trigger/Action	Connect to 5000+ apps via Zapier webhooks
Slack	Notification	Write	Handoff alerts, campaign updates, daily summaries
Webhook (Generic)	Custom	Push	Send call events to any HTTP endpoint

9.2 V2 Integrations

- Zenisth Transify — real-time multilingual translation during calls
- FlowraAI — deep workflow automation post-call
- Pipedrive, Zoho CRM
- Freshdesk, Zendesk (support ticket creation)
- DocraAI — auto-generate follow-up documents post-call

10. Roles & Permissions

Feature	Super Admin	Admin	Manager	Rep / SDR	Compliance Officer
Dashboard	✓ Full	✓ Full	✓ Full	✓ Limited	✓ Full (Read)
Knowledge Center	✓ Full	✓ Full	✓ Full	✗ No Access	✓ Read Only
Agent Builder	✓ Full	✓ Full	✓ View Only	✗ No Access	✓ Read Only
Campaign Manager	✓ Full	✓ Full	✓ Create/Edit	✓ View Only	✓ Read Only
Conversation Timeline	✓ Full	✓ Full	✓ Full	✓ Own Calls	✓ Full (Read)
Live Monitoring	✓ Full	✓ Full	✓ Full	✗ No Access	✓ Read Only
Human Handoff Console	✓ Full	✓ Full	✓ View + Override	✓ Accept Only	✓ Read Only
Compliance / Audit	✓ Full	✓ Read	✗ No Access	✗ No Access	✓ Full (Read + Flag)
Settings	✓ Full	✓ Full	✗ No Access	✗ No Access	✗ No Access
Billing	✓ Full	✓ View	✗ No Access	✗ No Access	✗ No Access
Delete Audit Logs	✓ Post-Retention Only	✗ Never	✗ Never	✗ Never	✗ Never

11. Master Error States & Edge Cases

11.1 System-Level Errors

Error	Trigger	User Message	System Action
Voice Infra Down	Twilio/Vonage API error	"Voice service temporarily unavailable. Your campaign is paused."	Auto-pause campaign. Alert admin. Retry queue after 5 mins.
STT Failure	Transcription service timeout	Silent failure — call continues. Flag added post-call.	Log error. Audio saved. Mark transcript as "Pending Review."
LLM Timeout (>3s)	LLM API response > 3 seconds	AI says: "Let me think for a moment..." — filler phrase plays.	Retry LLM with fallback prompt. If 2nd timeout: escalate.
Knowledge DB Unavailable	Vector DB timeout	AI escalates immediately.	Escalation triggered. Error logged. Admin notified.
Recording Storage Full	S3 bucket quota reached	"Recording unavailable for this call."	Alert admin. Pause new recordings. Existing calls unaffected.
Webhook Delivery Failure	CRM endpoint unreachable	Silent — queued for retry.	Retry 3x with exponential backoff. After 3 fails: flag in UI.

11.2 User-Caused Errors

Error	Trigger	User Message
Invalid CSV format	Missing required columns on upload	"CSV missing required columns: phone, name. Download template."
Unsupported file type	Wrong doc format in Knowledge Center	"Supported formats: PDF, DOCX, TXT, CSV. Please re-upload."
Agent not configured	Launching campaign with incomplete agent	"Agent setup incomplete. Complete: [missing fields] to continue."

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Duplicate campaign name	Campaign name already exists	"Campaign name already exists. Please use a unique name."
Calling hours violation (manual)	Admin tries to set window outside 8AM-9PM	"Calling outside 8AM–9PM local time is blocked for compliance."
Insufficient credits	Not enough balance for campaign size	"You need X credits. Available: Y. Top up here."

12. MVP vs V2 Roadmap

Feature	MVP	V2	V3
Outbound AI Calling	✓		
Inbound AI Support	✓		
Callback Scheduling Engine	✓		
Human Handoff Console	✓		
Conversation Timeline	✓		
Knowledge Center	✓		
Agent Builder (No-Code)	✓		
Live Monitoring Console	✓		
Compliance & Audit Layer	✓		
HubSpot + Salesforce Integration	✓		
Google Calendar + Calendly	✓		
Slack Notifications	✓		
Multi-language Voice		✓	
Transify Integration (Live Translation)		✓	
FlowraAI Workflow Orchestration		✓	
Custom LLM Fine-Tuning per Account		✓	
Sentiment Analysis Dashboard		✓	
Native Mobile App (iOS/Android)		✓	
Video Call Support			✓
Voice Cloning (Custom Brand Voice)			✓

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AI Coach (Trains reps from call data)			✓

13. Success Metrics & KPIs

Metric	Definition	MVP Target
Call Answer Rate	Calls answered / calls placed	> 25%
AI Resolution Rate	Calls resolved by AI without handoff	> 60%
Escalation Rate	Calls requiring human intervention	< 30%
Average Call Duration	Mean duration of all calls	< 4 minutes
Callback Conversion Rate	Callbacks that lead to meeting or sale	> 35%
DNC Compliance Rate	Calls correctly blocked per DNC list	100%
Transcript Accuracy	Word error rate in transcription	< 8% WER
Time to Handoff	Seconds from escalation trigger to rep alert	< 5 seconds
Source Attribution Coverage	AI responses with linked source	> 95%
Dashboard Load Time	Time to interactive on load	< 2 seconds

14. Open Questions & Decisions Needed

#	Question	Owner	Priority
1	Which telephony provider for MVP — Twilio, Vonage, or Telnyx?	Engineering	High
2	Which LLM for primary voice reasoning — GPT-4o or Claude Sonnet?	Product + Engineering	High
3	Pricing model — per minute, per call, or per agent seat?	Founder	High
4	Data residency requirements for enterprise customers?	Compliance + Engineering	Medium
5	Call recording consent laws by geography — how do we surface this?	Legal + Product	High
6	What is the default voice profile for AI agents — male, female, neutral?	Brand + Product	Medium
7	Should Zenisth Voice support SMS follow-up as a campaign action in MVP?	Product	Medium
8	Maximum concurrent calls per account on MVP infrastructure?	Engineering	High

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Automating conversations. Earning trust.

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